Bilinear and Quadratic Forms

An inner product does two jobs at once. It **pairs** two vectors and returns a scalar, linearly in each of them; and it **measures**, because $\langle v, v\rangle$ is a positive number whenever $v \neq 0$. Chapters 11 to 13 leaned on the second job constantly: lengths, angles, orthonormal bases, least squares, positive definiteness, and the singular value decomposition are all consequences of positivity. This chapter keeps the pairing and throws the measuring away. What is left is a **bilinear form**, a function $\beta\colon V \times V \to F$ linear in each slot, with no sign condition, no symmetry required, and no field restriction. The question of the chapter is what survives.

A surprising amount does. Forms still have matrices, and the matrices still change by a rule when the basis changes — but the rule is $\mathbf{A} \mapsto \mathbf{P}^{\top}\mathbf{A}\mathbf{P}$, not $\mathbf{A} \mapsto \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$. The organizing relation is therefore **congruence**, not similarity, and the two are genuinely different: congruent matrices need not share a single eigenvalue. What they do share is a rank, a symmetry type, and — over $\mathbb{R}$ — a triple of signs, which is Sylvester's law of inertia and the central classification theorem of the chapter. Beyond $\mathbb{R}$ the classification depends on the arithmetic of the field, and beyond symmetry there is a second world, the alternating forms, where every nondegenerate example looks like every other of the same size. Along the way we meet vectors on which a form vanishes without being zero, the groups that preserve a form, the geometry that physics runs on, and one algebra built to make a quadratic form into a square.

The field is a general F here. Since Chapter 10 the book has worked over $\mathbb{R}$ or $\mathbb{C}$, because inner products need them; this chapter opens the field back up, and that is the point rather than an afterthought. Chapters 11 to 13 turn out to have been the positive definite special case of one line of this theory. Two consequences of the general setting are worth flagging in advance. First, "positive" has no meaning over a field with no order, so the definiteness arguments of Chapter 13 simply cannot be run; what replaces them is either a square-class computation or nothing at all. Second, when $1 + 1 = 0$ — characteristic 2, Definition 0.7.2 — the identity $\beta(u, v) = \frac{1}{2}\big(q(u + v) - q(u) - q(v)\big)$ is unavailable, and symmetric and alternating forms stop being complementary. Every statement below that needs a characteristic hypothesis carries one, and every statement without one holds over every field.

What you need. From Chapter 7, multilinearity and alternation, which this chapter reuses rather than redefines (Definition 7.2.1, Definition 7.2.2, Definition 7.2.3, Theorem 7.2.1, Theorem 7.2.2), together with the determinant and its multiplicativity (Definition 7.4.1, Theorem 7.5.4, Theorem 7.4.5). From Chapter 2, the rank of a linear map (Definition 2.3.2). From Chapter 4, the matrix of a map, change of coordinates, and similarity (Definition 4.1.1, Theorem 4.1.1, Definition 4.2.1, Theorem 4.2.1, Theorem 4.2.3, Definition 4.2.2), together with equivalence of matrices and the rank normal form (Definition 4.3.1, Theorem 4.3.3). From Chapter 5, the dual space and its bases (Definition 5.1.1, Definition 5.1.2, Theorem 5.1.3, Corollary 5.1.4, Theorem 5.5.1, Theorem 5.3.2). From Chapter 0, fields and characteristic (Definition 0.7.1, Definition 0.7.2, Theorem 0.7.5), the transpose and symmetry (Definition 0.9.10, Definition 0.9.11, Theorem 0.9.5), and groups (Definition 0.10.1, Definition 0.10.2). From Chapter 1, the dimension counting that every classification here ends with: direct sums and complements (Theorem 1.7.4, Theorem 1.7.5, Theorem 1.7.6), the dimension formula for a sum of subspaces (Theorem 1.7.3), and the step from an inclusion to an equality (Theorem 1.6.8). From Chapter 3, the $\mathbf{L}\mathbf{D}\mathbf{L}^{\top}$ factorization (Theorem 3.6.8). From Chapters 11 to 13, everything used for the real and complex specializations: the orthogonal and unitary groups (Definition 11.7.2, Definition 11.7.3), the real spectral theorem and principal axes (Theorem 12.5.1, Corollary 12.5.2, Theorem 12.5.5), the orthogonal group in low dimension (Proposition 11.7.4, Theorem 12.6.1), and positivity (Definition 13.1.1, Theorem 13.1.4, Theorem 13.1.5, Proposition 13.1.2, Theorem 13.4.1). Those

last results are used **only** where the field is $\mathbb{R}$ or $\mathbb{C}$; they are never hypotheses of the general theory.

14.-1.1 Roadmap

- **Bilinear forms and their matrices.** What a form is, the Gram matrix that records it, the bijection between forms and matrices, and the form's companion map $V \to V^*$, whose kernel is the radical and whose invertibility is nondegeneracy.
- **Congruence.** Changing the basis replaces $\mathbf{A}$ by $\mathbf{P}^\top\mathbf{A}\mathbf{P}$. What that relation preserves — rank, symmetry type, nondegeneracy, the square class of the determinant — and what it destroys, starting with the eigenvalues.
- **Three kinds of form.** Symmetric, alternating, skew-symmetric, and Hermitian; why the middle two coincide exactly when the characteristic is not 2; polarization, and the quadratic form.
- **Completing the square.** Over any field of characteristic $\neq 2$, every symmetric form has an orthogonal basis. The proof is an induction, and the algorithm is symmetric row-and-column elimination.
- **Inertia.** Over $\mathbb{R}$ the number of positive, negative, and zero diagonal entries does not depend on the diagonalization. This classifies real symmetric matrices up to congruence and reconciles congruence with the spectral theorem.
- **Definiteness revisited.** Chapter 13's tests, restated as statements about signatures, and the second-derivative test they decide.
- **Where the form vanishes.** Isotropic vectors, totally isotropic subspaces, hyperbolic planes, and the Witt decomposition of a nondegenerate space.
- **Witt's theorems.** An isometry between subspaces extends to the whole space; one summand of an orthogonal sum may be canceled from both sides.
- **Symplectic forms.** A nondegenerate alternating form lives only in even dimension and has exactly one shape, over every field, with no characteristic hypothesis anywhere.
- **The classical groups.** $\mathrm{O}(n)$, $\mathrm{O}(p,q)$, $\mathrm{U}(n)$, and the symplectic group, all read off one equation $\mathbf{P}^\top\mathbf{A}\mathbf{P} = \mathbf{A}$.
- **Minkowski space.** The signature $(1,3)$ form, the light cone, hyperbolic rotations, and the triangle inequality running backwards.
- **A first look at Clifford algebras.** Making a quadratic form into a square, and how $\mathbb{C}$ and the quaternions come out of it.

14.-1.2 Named moves

- **Polarize.** A symmetric form and its quadratic function $q(v) = \beta(v,v)$ carry the same information, because β can be recovered from q by expanding $q(u+v)$ and dividing by 2. That division is the whole reason characteristic 2 is excluded whenever it is excluded.
- **Complete the square.** Every diagonalization theorem for symmetric forms in this chapter is this one manipulation, applied by induction on the dimension: isolate a variable that actually appears, absorb its cross terms into a perfect square, and recurse on what is left.
- **Split off a hyperbolic plane.** For indefinite forms this replaces "peel off an eigenvector". A single nonzero isotropic vector forces a two-dimensional piece of standard shape to split off orthogonally, and the induction runs on the rest.
- **Count signs, not values.** The numbers on a diagonalized form are not invariants; their signs are. Whenever a quantity refuses to be an invariant, ask whether some coarser feature of it is.
- **Congruence, not similarity.** At every theorem, ask which relation is in play. If the matrix is eaten twice by the same vector, the relation is congruence and eigenvalues are

not available; if it is eaten once and returns a vector, the relation is similarity and they are.